

SELEKTIVITATEA: GEOMETRIA

2018-2019 A2

$$\pi: x + y + z = 1 \rightarrow \vec{n}(1, 1, 1)$$

$$r: \begin{cases} x = 1 \\ y = t \\ z = t \end{cases} \rightarrow \vec{d}_r(0, 1, 1)$$

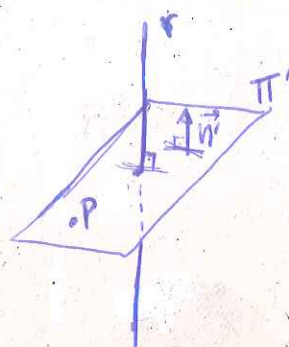
$$P(1, 1, 0)$$

a) $r \perp \pi'$ bide $\rightarrow \vec{n}' = \vec{d}_r(0, 1, 1)$

$$\begin{cases} P(1, 1, 0) \\ \vec{n}'(0, 1, 1) \end{cases}$$

Planoaren ekuazioa lortzeko: $Ax + By + Cz + D = 0 \rightarrow$

$$\rightarrow 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 + D = 0 \rightarrow D = -1 \Rightarrow \pi': y + z - 1 = 0$$

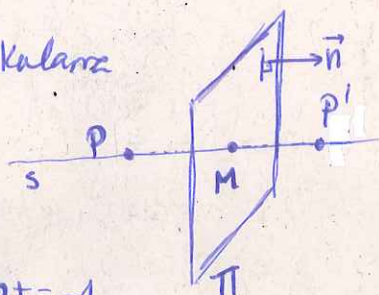


b) P-tik pasatzen den zuzene eta π planoarekiko perpendikularra

$$s: \begin{cases} x = 1 + t \\ y = 1 + t \\ z = t \end{cases}$$

s eta π -ren ebaki puntua

$$(1+t) + (1+t) + t = 1 \rightarrow 1+t+1+t+t=1 \rightarrow 3t=-1$$



$$t = -1/3$$

$$\rightarrow M \text{ ebaki-puntua } \begin{cases} x = 1 - 1/3 \\ y = 1 - 1/3 \\ z = -1/3 \end{cases} M(2/3, 2/3, -1/3)$$

$$\left(\frac{1+x'}{2}, \frac{1+y'}{2}, \frac{0+z'}{2} \right) = M(2/3, 2/3, -1/3)$$

$$\frac{1+x'}{2} = 2/3$$

$$x' = 1/3$$

$$\frac{1+y'}{2} = 2/3$$

$$y' = 1/3$$

$$\frac{z'}{2} = -1/3$$

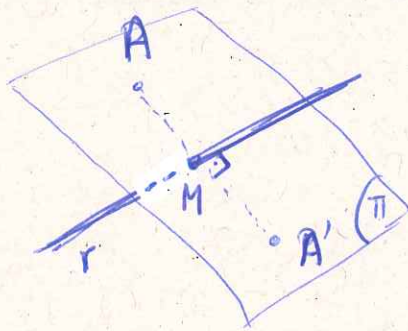
$$z' = -2/3$$

$$P'(1/3, 1/3, -2/3)$$

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$$A(-3, 1, -7)$$

$$r: \begin{cases} x = -1 + t \\ y = 3 + 2t \\ z = -1 + 2t \end{cases} \rightarrow \vec{d}_r(1, 2, 2)$$



$$\textcircled{1} \pi: \begin{cases} A(-3, 1, -7) \\ \vec{n}_\pi \equiv \vec{d}_r \equiv (1, 2, 2) \end{cases}$$

$$(x+3)1 + (y-1)2 + (z+7)2 = 0 \rightarrow x+3 + 2y-2 + 2z+14 = 0$$

$$\pi: \boxed{x + 2y + 2z + 15 = 0}$$

$$\textcircled{2} r \text{ eta } \pi\text{-ren ebaki-puntue bilatu} \rightarrow \textcircled{M}$$

$$r \text{ ordetkatu } \pi\text{-ren ekuazioan } -1+t + 2(3+2t) + 2(-1+2t) + 15 = 0$$

$$-1+t+6+4t-2+4t+15=0 \rightarrow 9t+18=0 \rightarrow t=-2$$

$$M: \begin{cases} x = -1-2 \\ y = 3-4 \\ z = -1+2(-2) \end{cases} \rightarrow M(-3, -1, -5)$$

$$\textcircled{3} P' \text{ bilatu } \frac{A+A'}{2} = M$$

$$\frac{-3+x'}{2} = -3 \rightarrow x' = -3$$

$$\frac{1+y'}{2} = -1 \rightarrow y' = -3$$

$$\frac{-7+z'}{2} = -5 \rightarrow z' = -3$$

$$\boxed{A'(-3, -3, -3)}$$

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$$A(3,3,3) \quad B(2,3,4) \quad C(0,0,4) \quad D(3,0,1)$$

a) A, B, C denotat plane Kalkulatu

$$\pi \begin{cases} A(3,3,3) \geq \vec{u}(-1,0,1) \\ B(2,3,4) \geq \vec{v}(2,3,0) \\ C(0,0,4) \geq \vec{v}(2,3,0) \end{cases} \quad \pi \begin{cases} x=0 & -\lambda + 2\mu = -\lambda + 2\mu \\ y=0 & +3\mu = 3\mu \\ z=4+\lambda & = 4+\lambda \end{cases}$$

$$D \in \pi ? \quad \begin{cases} 3 = -\lambda + 2\mu \rightarrow 3 = -(-3) + 0 \rightarrow 3 = 3 \checkmark \\ 0 = 3\mu \rightarrow \mu = 0 \\ 1 = 4 + \lambda \rightarrow \lambda = -3 \end{cases}$$

$D \in \pi \Rightarrow$ Beaz, Din punctele plane badienam dator.

b) $P(a,a,8) \in r(A,C)$

$$r: \begin{cases} A(3,3,3) \\ C(0,0,4) \end{cases} \geq \vec{d}(3,3,-1) \quad r: \begin{cases} x = 3t \\ y = 3t \\ z = 4 - t \end{cases}$$

Per egitate $\begin{cases} a = 3t \\ a = 3t \\ 8 = 4 - t \end{cases} \rightarrow a = 3 \cdot (-4) \rightarrow \boxed{a = -12}$ itan bekar da,

$t = -4$ $P(-12, -12, 8)$

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$$P(2, -1, 2)$$

$$r \equiv \frac{x}{2} = \frac{y-3}{1} = \frac{z-1}{-1}$$

$$\left\{ \begin{array}{l} \pi? \end{array} \right.$$

r zuzeneatik

$$\left\{ \begin{array}{l} A(0, 3, 1) \\ \vec{d}(2, 1, -1) = \vec{v} \end{array} \right.$$

$$\vec{AP} = \vec{u}(2, -4, 1)$$

$$\pi: \left\{ \begin{array}{l} P(2, -1, 2) \\ \vec{u}(2, -4, 1) \\ \vec{v}(2, 1, -1) \end{array} \right.$$

$$\pi: \left\{ \begin{array}{l} x = 2 + 2\lambda + 2\mu \\ y = -1 - 4\lambda + \mu \\ z = 2 + \lambda - \mu \end{array} \right.$$

Ekuazio implikaturak lortzeko

$$\left| \begin{array}{ccc} 2 & 2 & x-2 \\ -4 & 1 & y+1 \\ 1 & -1 & z-2 \end{array} \right| = 0 \rightarrow 2z - 4 + 4x - 8 + 2y + 2 - x + 2 + 2y + 2 + 8z - 16 = 0$$
$$\pi: \underline{\underline{3x + 4y + 10z - 22 = 0}}$$

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$$A(0, 2, 3)$$

$$B(-1, 1, 1)$$

$$\vec{d}(1, 1, 2)$$

$$r: \begin{cases} x = t \\ y = 2 + t \\ z = 3 + 2t \end{cases}$$

$$\text{dist}(P, M) = \text{dist}(P, N)$$

$$M(1, 0, 1)$$

$$N(0, 4, 2)$$

$$P(x, y, z) \quad \sqrt{(1-x)^2 + (0-y)^2 + (1-z)^2} = \sqrt{(0-x)^2 + (4-y)^2 + (2-z)^2}$$

$$(1-x)^2 + y^2 + (1-z)^2 = x^2 + (4-y)^2 + (2-z)^2$$

$$(1-t)^2 + (2+t)^2 + (-2-2t)^2 = t^2 + (2-t)^2 + (-1-2t)^2$$

$$1 - 2t + t^2 + 4 + 4t + t^2 + 4 + 8t + 4t^2 = t^2 + 4 - 4t + t^2 + 1 + 4t + 4t^2$$

$$4 + 10t = 0 \rightarrow 10t = -4 \rightarrow t = -\frac{4}{10} \rightarrow t = -\frac{2}{5}$$

P punkt:

$$r: \begin{cases} x = -2/5 = -2/5 \\ y = 2 - 2/5 = 8/5 \\ z = 3 + 2 \cdot (-2/5) = \frac{15}{5} - \frac{4}{5} = \frac{11}{5} \end{cases}$$

$$\boxed{P\left(-\frac{2}{5}, \frac{8}{5}, \frac{11}{5}\right)}$$

$$a) \pi_1: x - 3y + z = 0$$

$$\pi_2: 2x - y + 3z - 5 = 0$$

$$P(2, 6, 5)$$

} Eskatzen den murena planoekiko paralela baldin bada $\Rightarrow \vec{d} \perp \vec{n}_1$ eta $\vec{d} \perp \vec{n}_2$ beteketa

$$\vec{n}_1(1, -3, 1)$$

$$\vec{n}_2(2, -1, 3)$$

$$\vec{d} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -3 & 1 \\ 2 & -1 & 3 \end{vmatrix} = -9\vec{i} + 2\vec{j} - \vec{k} + 6\vec{k} + \vec{i} - 3\vec{j} \Rightarrow$$

$$\Rightarrow \vec{d} = (-8, -1, 5)$$

$$r: \begin{cases} P(2, 6, 5) \\ \vec{d}(-8, -1, 5) \end{cases}$$

$$r: \begin{cases} x = 2 - 8t \\ y = 6 - t \\ z = 5 + 5t \end{cases}$$

$$\boxed{\frac{x-2}{-8} = \frac{y-6}{-1} = \frac{z-5}{5}}$$

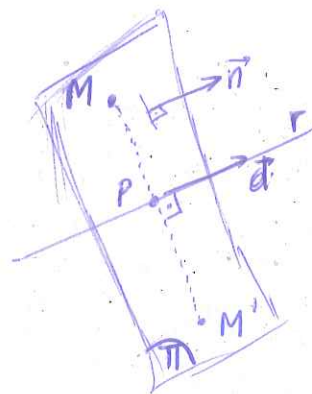
$$b) \text{dist}(r, \pi_1) = \text{dist}(P, \pi_1) = \frac{|2 - 3 \cdot 6 + 5|}{\sqrt{1^2 + 3^2 + 1^2}} = \frac{|-11|}{\sqrt{11}} \rightarrow \boxed{\text{dist}(r, \pi_1) = \sqrt{11} \text{ u}}$$

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$$M(1, -3, 7)$$

$$r \rightarrow A(1, -3, 4)$$

$$B(0, -4, 1)$$



$$\vec{d} = \vec{n}$$

r zuzenaren bektorea erabiliko dogu Π planaren ekuazioa lortzeko

$$r \text{ zuzenaren norabide bektorea} \rightarrow \vec{d} = \overrightarrow{AB}(-1, -1, -3)$$

$$r \perp \Pi \text{ denez} \rightarrow \vec{d} = \vec{n} = \begin{pmatrix} -1 \\ -1 \\ -3 \end{pmatrix}$$

A B C

① Π planaren ekuazioa:

$$Ax + By + Cz + D = 0$$

M puntua eta $\vec{n}$ bektorea ordenkatuz:

$$-1 \cdot 1 - 1 \cdot (-3) - 3 \cdot 7 + D = 0 \rightarrow -1 + 3 - 21 + D = 0 \rightarrow \boxed{D = 19}$$

$$\Pi: -x - y - 3z + 19 = 0 \text{ edo } \boxed{\Pi: x + y + 3z - 19 = 0}$$

② r zuzenaren eta Π planaren arteko ebaki puntua $\rightarrow P$

$$r: \begin{cases} x = -t \\ y = -4 - t \\ z = 1 - 3t \end{cases} \rightarrow \Pi \text{ planaren ekuazioa ordenkatuz:}$$

$$-t + (-4 - t) + 3(1 - 3t) - 19 = 0 \rightarrow -t - 4 - t + 3 - 9t - 19 = 0$$

$$-11t = 20 \rightarrow \boxed{t = -20/11}$$

Ebaki puntua:

$$\begin{cases} x = -(-20/11) \\ y = -4 - (-20/11) \\ z = 1 - 3(-20/11) \end{cases} \rightarrow \boxed{P\left(\frac{20}{11}, -\frac{24}{11}, \frac{71}{11}\right)}$$

Ebaki puntua
 $r \cap \Pi$

③ $M' \rightarrow M$ -ren simetria: $\frac{M + M'}{2} = P$

$M'(x', y', z')$ izanik:

$$\frac{1 + x'}{2} = \frac{20}{11}$$

$$x' = \frac{29}{11}$$

$$\frac{-3 + y'}{2} = -\frac{24}{11}$$

$$y' = -\frac{15}{11}$$

$$\frac{7 + z'}{2} = \frac{71}{11}$$

$$z' = \frac{65}{11}$$

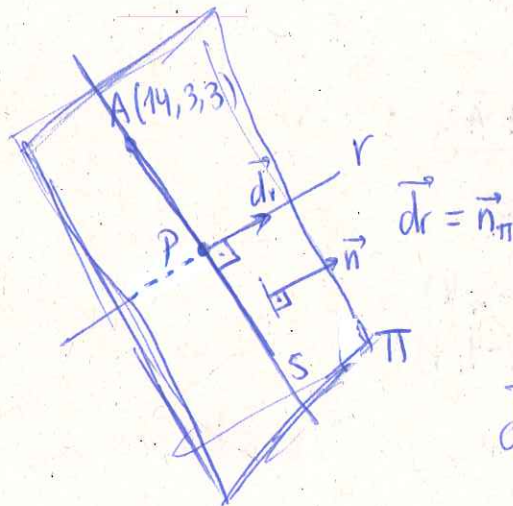
$$\boxed{M'\left(\frac{29}{11}, -\frac{15}{11}, \frac{65}{11}\right)}$$

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$$r: \frac{x}{2} = \frac{y-3}{-2} = \frac{z-1}{3}$$

$$A(14, 3, 3) \in S$$

$$r \perp S$$



$$\vec{d}_r = \vec{n}_{\pi} \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}_{A \ B \ C}$$

① A-tik pasatzen den eta $r \perp \pi$ den π planua kalkulatu:

$$\pi: Ax + By + Cz + D = 0$$

$$2 \cdot 14 - 2 \cdot 3 + 3 \cdot 3 + D = 0 \rightarrow 28 - 6 + 9 + D = 0 \rightarrow D = -31$$

$$\pi: 2x - 2y + 3z - 31 = 0$$

② $r \cap \pi$ ebaki-puntua $\rightarrow P$

$$r: \begin{cases} x = 2t \\ y = 3 - 2t \\ z = 1 + 3t \end{cases}$$

$$2(2t) - 2(3 - 2t) + 3(1 + 3t) - 31 = 0$$

$$4t - 6 + 4t + 3 + 9t - 31 = 0$$

$$17t - 34 = 0 \rightarrow 17t = 34 \rightarrow t = 2$$

$$P \text{ Ebaki-puntua } \begin{cases} x = 2 \cdot 2 \\ y = 3 - 2 \cdot 2 \\ z = 1 + 3 \cdot 2 \end{cases}$$

$$\underline{P(4, -1, 7)}$$

③ S zuzenaren ekuazioa

$$A(14, 3, 3)$$

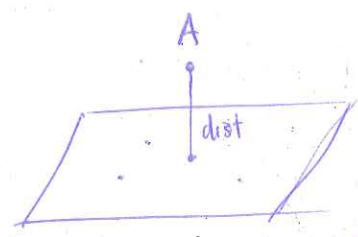
$$P(4, -1, 7)$$

$$\vec{d}_S(10, 4, -4)$$

$$S: \frac{x-14}{10} = \frac{y-3}{4} = \frac{z-3}{-4}$$

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$$\left. \begin{array}{l} A(4,4,3) \\ B(1,1,0) \\ C(1,0,1) \\ D(0,1,1) \end{array} \right\} \Pi \text{ planoan}$$



dist(A, Π) ?

① Π Planaren ekuazio berba:

$$\begin{array}{l} B(1,1,0) \rightarrow \vec{u}(0,1,-1) \\ C(1,0,1) \rightarrow \vec{v}(1,-1,0) \\ D(0,1,1) \end{array}$$

Ekuazio parametrikak:

$$\Pi: \begin{cases} x = 1 + \mu \\ y = 1 + \lambda - \mu \\ z = -\lambda \end{cases} \quad \begin{matrix} \vec{u} \\ \vec{v} \end{matrix}$$

Ekuazio implizitua:

$$\begin{vmatrix} 0 & 1 & x-1 \\ 1 & -1 & y-1 \\ -1 & 0 & z \end{vmatrix} = 0 \Rightarrow -(y-1) - (x-1) - z = 0$$

$$-y+1-x+1-z=0 \rightarrow \boxed{x+y+z-2=0} \leftarrow \Pi$$

② Distantzia kalkulatu:

$A(4,4,3)$

$$\boxed{\text{dist}(A, \Pi) = \frac{|4+4+3-2|}{\sqrt{1^2+1^2+1^2}} = \frac{|9|}{\sqrt{3}} = \underline{\underline{3\sqrt{3} \text{ u}}}}$$

$$r \begin{cases} P(1,2,3) \\ Q(-1,0,1) \end{cases} \rightarrow \vec{dr} (2,2,2)$$

$$a) \pi_1 \begin{cases} r \perp \pi_1 \rightarrow \vec{dr} = \vec{n}_{\pi_1} = (2,2,2) \\ A(4,-2,-1) \end{cases} \begin{matrix} A & B & C \\ x & y & z \end{matrix}$$

$$\pi_1: Ax + By + Cz + D = 0 \rightarrow 2 \cdot 4 + 2 \cdot (-2) + 2 \cdot (-1) + D = 0 \rightarrow$$

$$\rightarrow 8 - 4 - 2 + D = 0 \rightarrow D = -2$$

$$\pi_1: \boxed{2x + 2y + 2z - 2 = 0}$$

EHU
(Soluzionarioan txarto)

$$b) \pi_2 \begin{cases} r \perp \pi_2 \rightarrow \vec{dr} = \vec{n}_{\pi_2} (2,2,2) \\ B(2,1,-3) \end{cases}$$

$$\pi_2: Ax + By + Cz + D = 0 \rightarrow 2 \cdot 2 + 2 \cdot 1 + 2 \cdot (-3) + D = 0 \rightarrow$$

$$\rightarrow 4 + 2 - 6 + D = 0 \rightarrow \boxed{D = 0}$$

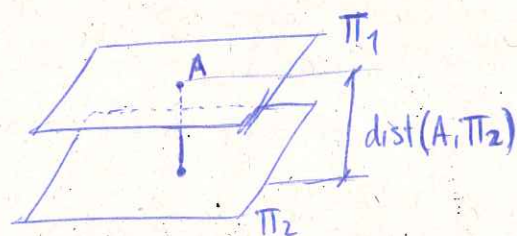
$$\boxed{\pi_2: 2x + 2y + 2z = 0}$$

(Soluzionarioan txarto)

$$c) \text{dist}(\pi_1, \pi_2) \quad \text{Planak paralelak dira.}$$

$$A \in \pi_1 \rightarrow \text{dist}(\pi_1, \pi_2) = \text{dist}(A, \pi_2)$$

$$\text{dist}(A, \pi_2) = \frac{|2 \cdot 4 + 2 \cdot (-2) + 2 \cdot (-1)|}{\sqrt{2^2 + 2^2 + 2^2}}$$



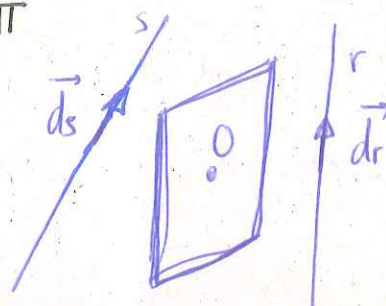
$$\boxed{\text{dist}(A, \pi_2) = \frac{|2|}{\sqrt{12}} = \frac{\sqrt{3}}{3} u}$$

(EHU-ko soluzionarioan txarto dago ariketa hau)

$\Pi // r$ $\Pi // s$ $(0,0,0) \in \Pi$

$$r: \frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-1}{1} \rightarrow \vec{dr}(1, -1, 1)$$

$$s: \begin{cases} A(0, 1, 1) \\ B(1, 1, 0) \end{cases} \rightarrow \vec{ds}(1, 0, -1)$$



Π planoaren bi bektore $\vec{dr}$ eta $\vec{ds}$ izan daitezke, planoarekiko paraleloak direlako.

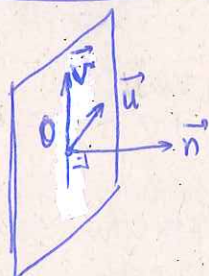
$$\Pi: \begin{cases} O(0,0,0) \\ \vec{u} = \vec{dr}(1, -1, 1) \\ \vec{v} = \vec{ds}(1, 0, -1) \end{cases}$$

$$\Pi: \begin{cases} x = \lambda + \mu \\ y = -\lambda \\ z = \lambda - \mu \end{cases}$$

Planoaren ekuazio implizitua eskatzen badenakue:

$$\begin{vmatrix} 1 & 1 & x-0 \\ -1 & 0 & y-0 \\ 1 & -1 & z-0 \end{vmatrix} = 0 \rightarrow 0 + x + y + y + z = 0 \rightarrow \boxed{x + 2y + z = 0}$$

Beste modu bat



$\vec{u}$ eta $\vec{v}$ -ren biderketa bektorialaren bidez, planoaren bektore normala ($\vec{n}$) lortu daiteke.

$$\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{vmatrix} = \vec{i} + \vec{j} + \vec{k} + \vec{j} = \vec{i} + 2\vec{j} + \vec{k} \rightarrow \vec{n} = (1, 2, 1)$$

Ekuazio implizituan 0 puntua eta $\vec{n}$ ordezkatuz $\rightarrow D$ lortzen da

$$Ax + By + Cz + D = 0 \rightarrow 1 \cdot 0 + 2 \cdot 0 + 1 \cdot 0 + D = 0 \rightarrow \boxed{D = 0}$$

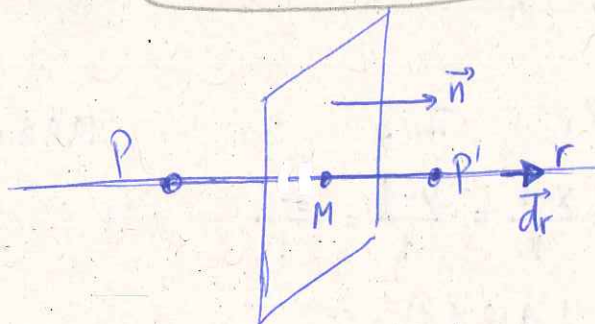
Beraz ekuazio implizitua

$$\boxed{x + 2y + z = 0}$$

$$\Pi: x - 3y + 2z = 7$$

a) $P(3, -8, 4) \rightarrow P'?$

b) $\text{dist}(P, P')$



a) $\vec{dr} = \vec{n} = (1, -3, 2)$

$$r: \begin{cases} P(3, -8, 4) \\ \vec{dr}(1, -3, 2) \end{cases} \Rightarrow r: \begin{cases} x = 3 + t \\ y = -8 - 3t \\ z = 4 + 2t \end{cases}$$

↳ M puntua $\rightarrow$ r eta Π -ren arteko ebaki puntua izango da.
r-ren adierazpenak Π planaren ekuazioan ordezkatuz:

$$\Pi: (3+t) - 3(-8-3t) + 2(4+2t) = 7$$

$$3+t + 24 + 9t + 8 + 4t = 7$$

$$14t = -28 \rightarrow t = -2$$

r-ren ekuazio parametrikotik ordezkatuz $\Rightarrow$ (M)

$$M \rightarrow \begin{cases} x = 3 - 2 \\ y = -8 - 3(-2) \\ z = 4 + 2(-2) \end{cases} \rightarrow \underline{M(1, -2, 0)}$$

P puntuaren simetrikoa M-rekiko (edo Π planarekiko):

$$\Rightarrow \underline{P'(-1, 4, -4)}$$

$$\left. \begin{aligned} 1 &= \frac{3+x'}{2} \\ -2 &= \frac{-8+y'}{2} \\ 0 &= \frac{4+z'}{2} \end{aligned} \right\} \Rightarrow$$

b) Bi punturen arteko distantzia $\Rightarrow$ Beaien artean definitutako bektorearen moduluen bardina da

$$\text{dist}(P, P') = |\vec{PP'}| = \sqrt{(-4)^2 + 12^2 + (-8)^2}$$

$$P \xrightarrow{\vec{PP'}} P'$$

$$\vec{PP'} = (-4, 12, -8)$$

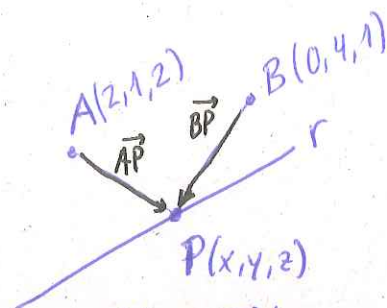
$$\underline{\underline{\text{dist}(P, P') = 4\sqrt{14} \text{ u}}}$$

$$A(2,1,2)$$

$$B(0,4,1)$$

$$r: x = y - 2 = \frac{z-3}{2} \rightarrow \vec{d}_r(1,1,2)$$

$$C(0,2,3)$$



$$|\vec{AP}| = |\vec{BP}|$$

izan behar da.

a) $P \in r$

A eta B puntuetatik distantzia berdinera.

r zuzena parametrikotara pasatuta
sinpleagoa eta erosoagoa da.

C eta $\vec{d}_r$ erabiliz, ekuazio parametrikok honek
izango dira:

$$r: \begin{cases} x = t \\ y = 2+t \\ z = 3+2t \end{cases}$$

r zuzeneko edozein puntuk ekuazio honek
bete behar dauzala kontuan hartu (baita
P puntuak ere)

$$\vec{AP} = (x-2, y-1, z-2) \rightarrow \vec{AP} = (t-2, 2+t-1, 3+2t-2)$$

$$\vec{BP} = (x, y-4, z-1) \rightarrow \vec{BP} = (t, 2+t-4, 3+2t-1) \Rightarrow$$

$$\Rightarrow \begin{cases} \vec{AP} = (t-2, 1+t, 1+2t) \\ \vec{BP} = (t, t-2, 2+2t) \end{cases}$$

$$|\vec{AP}| = |\vec{BP}| \text{ izan behar denez} \Rightarrow \sqrt{(t-2)^2 + (1+t)^2 + (1+2t)^2} = \sqrt{t^2 + (t-2)^2 + (2+2t)^2}$$

$$\rightarrow (t-2)^2 + (1+t)^2 + (1+2t)^2 = t^2 + (t-2)^2 + (2+2t)^2 \rightarrow$$

$$\rightarrow 1+2t+t^2 + 1+4t+4t^2 = t^2 + 4+8t+4t^2 \rightarrow 2+6t = 4+8t \rightarrow$$

$$\rightarrow -2t = 2 \rightarrow \boxed{t = -1}$$

Beraz, gu bilatzen gogozan P puntua (A eta B puntuetatik distantzia berdinera)

$$r: \begin{cases} x = -1 \\ y = 2-1 \\ z = 3+2 \cdot (-1) \end{cases} \rightarrow \boxed{P(-1, 1, 1)}$$

$$b) \quad \Pi \begin{cases} A \in \Pi \\ r \perp \Pi \end{cases} \rightarrow \vec{dr} = \vec{n} \begin{matrix} A & B & C \\ (2, 1, 2) \end{matrix} \begin{matrix} x & y & z \\ (1, 1, 2) \end{matrix}$$



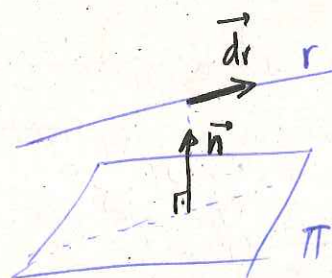
Plano baten ekuazio implizitua (orokorrean): $Ax + By + Cz + D = 0$

$$2 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 + D = 0 \rightarrow D = -7$$

Beraz, Π planoaren ekuazioa: $\Pi: \underline{x + y + 2z - 7 = 0}$

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$$a) \quad r: \begin{cases} 3x + y - z = 2 \\ 2x + y + 4z = 1 \end{cases}$$



Paralelaak

$$\Pi: 2x + (a+1)(y-3) + a(z-1) = 0$$

$$\Pi // r \Rightarrow a?$$

$$\Pi // r \text{ badira} \rightarrow \vec{n} \perp \vec{dr} \text{ izan behar da.} \Rightarrow \underline{\vec{n} \cdot \vec{dr} = 0}$$

$$\circledast \quad \Pi: 2x + ay - 3a + y - 3 + az - a = 0 \rightarrow \begin{matrix} A & B & C \\ 2x + (a+1)y + az - 4a - 3 = 0 \end{matrix}$$

Beraz, Π planoaren bektore normalak itxura hau dauka:

$$\underline{\vec{n} (2, a+1, a)}$$

$\circledast$ r zuzenaren norabide-bektoreak lortzeko $\rightarrow r$ parametrikotara

$$\text{pasatuz egin darteke} \Rightarrow r: \begin{cases} 3x + y - z = 2 \\ 2x + y + 4z = 1 \end{cases} \quad \text{Adibidez } \underline{z = t} \text{ eginez}$$

$$r: \begin{cases} 3x + y - t = 2 \\ 2x + y + 4t = 1 \end{cases} \rightarrow \underline{y = 2 + t - 3x} \rightarrow 2x + (2 + t - 3x) + 4t = 1 \rightarrow -x + 2 + 5t = 1 \rightarrow$$

$$\rightarrow \underline{x = 1 + 5t}$$

$$y = 2 + t - 3(1 + 5t) \rightarrow \underline{y = -1 - 14t}$$

$$\Rightarrow r: \begin{cases} x = 1 + 5t \\ y = -1 - 14t \\ z = t \end{cases}$$

$$\underline{\vec{dr} (5, -14, 1)}$$

$$\circledast \quad \vec{n} \cdot \vec{dr} = 0 \rightarrow (2, a+1, a) \cdot (5, -14, 1) = 0$$

$$10 - 14(a+1) + a = 0 \rightarrow 10 - 14a - 14 + a = 0 \rightarrow \underline{a = -\frac{4}{13}}$$

b) $P(1, 0, -3)$

$P \in \Pi$?

$a = -\frac{4}{13}$ izanik, planoaren ekuazioa honela geratuko litzateke:

$$\Pi: 2x + \left(-\frac{4}{13} + 1\right)y + \left(-\frac{4}{13}\right)z - 4 \cdot \frac{-4}{13} - 3 = 0$$

$$2x + \frac{9}{13}y - \frac{4}{13}z - \frac{23}{13} = 0 \xrightarrow{\cdot 13} \boxed{26x + 9y - 4z - 23 = 0}$$

$P(1, 0, -3)$ planoaren ekuazioan ordezkatzu:

$$26 \cdot 1 + 9 \cdot 0 - 4(-3) - 23 = 0 \rightarrow 26 + 12 - 23 = 0 \rightarrow \boxed{15 \neq 0}$$

Beraz, P puntuak ez da Π planokoa

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a) $P(-1, 2, 3)$

$\Pi \begin{cases} \vec{a}(-1, -2, -3) \\ \vec{b}(1, 3, 5) \end{cases}$ planoarekiko paraleloak badira $\rightarrow$ plano bektore moduan hartu daitezke.

Planoaren ekuazio parametrikokoak:

$$\Pi: \begin{cases} x = -1 - \lambda + \mu \\ y = 2 - 2\lambda + 3\mu \\ z = 3 - 3\lambda + 5\mu \end{cases}$$

Ekuazio inplizitua lortzeko:

$$\begin{array}{cc} \vec{a} & \vec{b} \\ \downarrow & \downarrow \end{array} \begin{vmatrix} -1 & 1 & x+1 \\ -2 & 3 & y-2 \\ -3 & 5 & z-3 \end{vmatrix} = 0 \Rightarrow$$

$$\Rightarrow -3(z-3) - 10(x+1) - 3(y-2) + 9(x+1) + 5(y-2) + 2(z-3) = 0$$

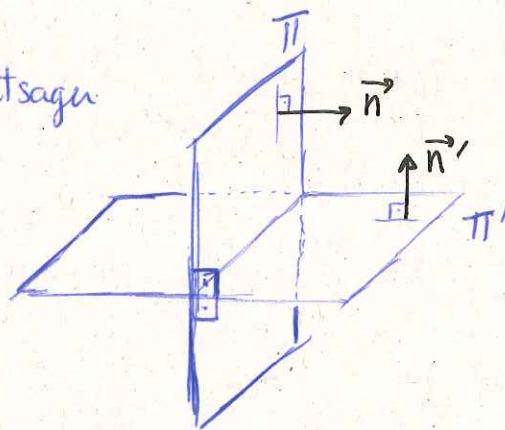
$$-3z + 9 - 10x - 10 - 3y + 6 + 9x + 9 + 5y - 10 + 2z - 6 = 0$$

$$\Pi: \boxed{-x + 2y - z - 2 = 0}$$

b) $mx - y + 5z = 8 \leftarrow \Pi'$ planoaren deituko deitua

$\Pi' \perp \Pi$ izateko $\rightarrow m?$

$\Pi' \perp \Pi \Rightarrow \vec{n}' \perp \vec{n} \Rightarrow \boxed{\vec{n}' \cdot \vec{n} = 0}$



$\vec{n}'(m, -1, 5)$

$\vec{n}(-1, 2, -1)$

Bakoitzaren bere ekuazioak.

$\vec{n}' \cdot \vec{n} = 0 \rightarrow (m, -1, 5) \cdot (-1, 2, -1) = 0 \rightarrow -m - 2 - 5 = 0 \rightarrow$

$\rightarrow \boxed{m = -7}$ izan behar da

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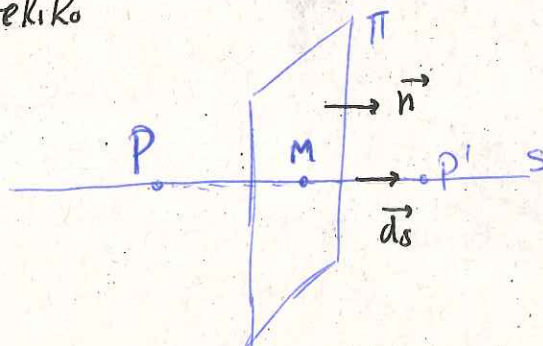
r: $\begin{cases} \vec{v}(1, 2, 3) \\ P' \in r \end{cases} \Rightarrow \vec{dr}$

$\rightarrow P'$ -ren koordenatuak lortu eta r-ren ekuazioa idatzi.

P' puntua P -ren simetrikoa da Π planarekiko

$P(0, -2, 0)$

$\Pi: x + 3y + z = 5 \rightarrow \vec{n}(1, 3, 1)$



s: $\begin{cases} \vec{ds} = \vec{n}(1, 3, 1) \text{ izango da} \\ P(0, -2, 0) \end{cases}$

Beraz, s zuzenaren ekuazio parametrikak:

s: $\begin{cases} x = t \\ y = -2 + 3t \\ z = t \end{cases}$

Π planoan $\rightarrow$ s eta Π -ren arteko ebaki puntua $\rightarrow (M)$
 $t + 3(-2 + 3t) + t = 5 \rightarrow t - 6 + 9t + t = 5$

$11t = 11 \rightarrow \boxed{t = 1}$

$\boxed{M(1, 1, 1)}$

$1 = \frac{0 + x'}{2} \rightarrow x' = 2$

$1 = \frac{-2 + y'}{2} \rightarrow y' = 4 \rightarrow P'(2, 4, 2)$

$1 = \frac{0 + z'}{2} \rightarrow z' = 2$

r: $\begin{cases} P'(2, 4, 2) \\ \vec{dr}(1, 2, 3) \end{cases}$

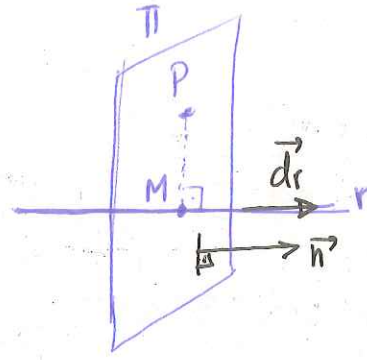
$\Rightarrow$ Ekuazio jarraitua:

$\boxed{\frac{x-2}{1} = \frac{y-4}{2} = \frac{z-2}{3}}$

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$P(2, -1, 3)$

$r: \frac{x}{3} = \frac{y+7}{5} = \frac{z-2}{2} \rightarrow \vec{dr}(3, 5, 2)$



(M) → P puntuaren proiektzioa r-ren gainean
↓
r eta Pi-ren ebaki puntua ere bada

a) Pi plano laguntzarilea definituko
dogu lehenengo → $r \perp \Pi$ izanik
 $\vec{dr} = \vec{n}(3, 5, 2)$

Pi-ren ekuazio implizitua $\begin{cases} P(2, -1, 3) \\ \vec{n}(3, 5, 2) \end{cases} \Rightarrow Ax + By + Cz + D = 0 \rightarrow$
 $\rightarrow 3 \cdot 2 + 5 \cdot (-1) + 2 \cdot 3 + D = 0 \rightarrow 6 - 5 + 6 + D = 0 \rightarrow D = -7$

$\Pi: 3x + 5y + 2z - 7 = 0$

M → r eta Pi-ren arteko ebaki puntua ere bada:

r parametrizatzen $\begin{cases} x = 3t \\ y = -7 + 5t \\ z = 2 + 2t \end{cases}$ Pi planean orderkatuz:
 $3 \cdot 3t + 5(-7 + 5t) + 2(2 + 2t) - 7 = 0 \rightarrow$
 $\rightarrow 9t - 35 + 25t + 4 + 4t - 7 = 0 \rightarrow 38t - 38 = 0 \rightarrow t = 1$

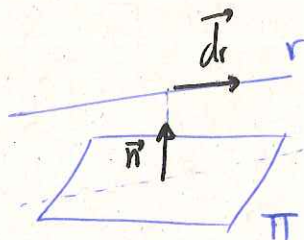
Beraz, r-en orderkatuz $\Rightarrow$ (M) $\begin{cases} x = 3 \cdot 1 \\ y = -7 + 5 \cdot 1 \\ z = 2 + 2 \cdot 1 \end{cases}$ P puntuaren proiektzioa r-ren gainean:
 $\boxed{M(3, -2, 4)}$

b) $\text{dist}(P, r) = \text{dist}(P, M) = |\vec{PM}| = \sqrt{1^2 + (-1)^2 + 1^2} \rightarrow \boxed{\text{dist}(P, r) = \sqrt{3} \text{ u}}$
 $\vec{PM} = (1, -1, 1)$

c) $P(2, -1, 3)$ $(M = \frac{P+P'}{2})$
 $M(3, -2, 4)$ $3 = \frac{2+x'}{2}$
 $P'(x', y', z')$ $-2 = \frac{-1+y'}{2} \Rightarrow \boxed{P'(4, -3, 5)}$
 $4 = \frac{3+z'}{2}$

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$$r: \begin{cases} 2x - y + z = 0 \\ x - y + 4z = 1 \end{cases}$$



paraleloak

$$\pi: 3x - 5y + Az = -31 \rightarrow \underline{\underline{\vec{n}(3, -5, A)}}$$

a) $\pi // r$ izateko $\rightarrow A$?

$$\pi // r \text{ badira} \rightarrow \vec{n} \perp \vec{dr} \rightarrow \boxed{\vec{n} \cdot \vec{dr} = 0}$$

$\vec{dr}$ lortzeko $\Rightarrow$ Zuzenaren ekuazio implizituetan $\boxed{z=t}$ eginez, ekuazio parametrikokoak lortzen dira $\rightarrow$ Hortik $\vec{dr}$

$$r: \begin{cases} 2x - y + t = 0 \\ x - y + 4t = 1 \end{cases} \rightarrow \boxed{2x + t = y}$$

$$x - (2x + t) + 4t = 1 \rightarrow x - 2x - t + 4t = 1$$

$$-x = 1 - 3t \rightarrow \boxed{x = -1 + 3t}$$

$$y = 2(-1 + 3t) + t \rightarrow y = -2 + 6t + t \rightarrow \boxed{y = -2 + 7t}$$

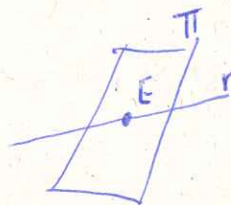
r zuzenaren ekuazio parametrikokoak:

$$r: \begin{cases} x = -1 + 3t \\ y = -2 + 7t \\ z = t \end{cases} \Rightarrow \underline{\underline{\vec{dr}(3, 7, 1)}}$$

$$\boxed{\vec{n} \cdot \vec{dr} = 0} \rightarrow (3, -5, A) \cdot (3, 7, 1) = 0 \rightarrow 9 - 35 + A = 0$$

zuzendua

$$\boxed{A = 26}$$
 izan behar da.



b) $A = 12 \rightarrow \pi: 3x - 5y + 12z = -31$

r -ren parametrikokoak π planon ordezkaturaz $\Rightarrow E$ ebaki-puntua

$$\pi: 3(-1 + 3t) - 5(-2 + 7t) + 12t = -31 \rightarrow -3 + 9t + 10 - 35t + 12t = -31$$

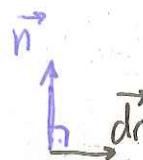
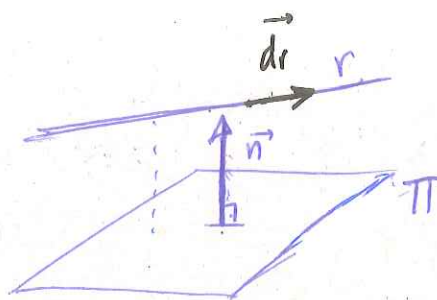
$$-38t = -38 \rightarrow \boxed{t=1} \Rightarrow r\text{-ren ekuazio parametrikoei}$$

$$\begin{cases} x = -1 + 3 \cdot 1 \\ y = -2 + 7 \cdot 1 \\ z = 1 \end{cases}$$

Ebaki-puntua $\underline{\underline{E(2, 5, 1)}}$

$$r: \begin{cases} 4x - 3y + 4z = -1 \\ 3x - 2y + z = -3 \end{cases}$$

$$\pi: 2x - y + Az = 0 \rightarrow \vec{n}(2, -1, A)$$



a) $r // \pi \rightarrow A?$

$$r // \pi \rightarrow \vec{dr} \perp \vec{n} \rightarrow \boxed{\vec{dr} \cdot \vec{n} = 0}$$

$\vec{dr}$ lortzeko $\rightarrow r$ zurena ekuazio parametrikotara jarri.

$z = t$ eginez

$$r: \begin{cases} 4x - 3y + 4t = -1 \\ 3x - 2y + t = -3 \end{cases} \xrightarrow{\begin{matrix} \cdot 2 \\ -3 \end{matrix}} \begin{cases} 8x - 6y + 8t = -2 \\ 9x - 6y + 3t = -9 \end{cases}$$

$$\begin{matrix} -x & +5t = 7 \end{matrix} \rightarrow \boxed{x = 5t - 7}$$

$$3(5t - 7) - 2y + t = -3$$

$$15t - 21 - 2y + t = -3 \rightarrow 15t - 21 + t + 3 = 2y$$

$$16t - 18 = 2y \rightarrow \boxed{y = -9 + 8t}$$

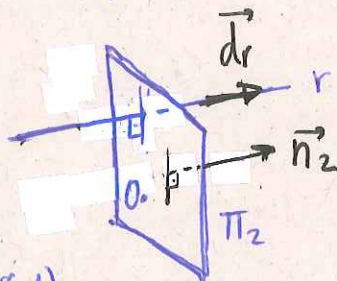
Beraz,

$$r: \begin{cases} x = -7 + 5t \\ y = -9 + 8t \\ z = t \end{cases} \rightarrow \boxed{\vec{dr}(5, 8, 1)}$$

$$\vec{dr} \cdot \vec{n} = 0 \rightarrow (5, 8, 1) \cdot (2, -1, A) = 0 \rightarrow 10 - 8 + A = 0 \rightarrow \boxed{A = -2}$$

b) $\pi_2 \perp r$

$$O(0, 0, 0) \in \pi_2$$



$$\vec{dr} = \vec{n}_2(5, 8, 1)$$

$$\pi_2 \begin{cases} \pi_2 \perp r \rightarrow \vec{n}_2 = \vec{dr}(5, 8, 1) \\ O(0, 0, 0) \end{cases} \xrightarrow{A \ B \ C} \begin{matrix} x \ y \ z \end{matrix}$$

$$Ax + By + Cz + D = 0 \rightarrow 0 + 0 + 0 + D = 0$$

$$\boxed{D = 0}$$

Beraz, π_2 planoak: $\boxed{5x + 8y + z = 0}$

$$r: \frac{x-2}{2} = \frac{y+1}{3} = \frac{z-2}{2} \quad / \quad \text{Per}$$

$$\left. \begin{array}{l} \pi_1: 3x + 4y - 1 = 0 \\ \pi_2: 4x - 3y + 9 = 0 \end{array} \right\} \text{dist}(P, \pi_1) = \text{dis}(P, \pi_2)$$

P? $P(x, y, z)$

r zuzena parametroetan idatziz:

$$\begin{cases} x = 2 + 2t \\ y = -1 + 3t \\ z = 2 + 2t \end{cases}$$

Distantziakidetasuna ren baldintza erabiliz: $\text{dist}(P, \pi_1) = \text{dist}(P, \pi_2)$

$$\frac{|3x + 4y - 1|}{\sqrt{3^2 + 4^2 + 0^2}} = \frac{|4x - 3y + 9|}{\sqrt{4^2 + (-3)^2 + 0^2}} \quad \leftarrow \begin{array}{l} \text{r zuzenaren} \\ \text{ekuazio parametroekin} \\ \text{ordetkatuz} \end{array}$$

$$\frac{|3(2+2t) + 4(-1+3t) - 1|}{\sqrt{9+16}} = \frac{|4(2+2t) - 3(-1+3t) + 9|}{\sqrt{16+9}}$$

$$|6 + 6t - 4 + 12t - 1| = |8 + 8t + 3 - 9t + 9| \Rightarrow |18t + 1| = |20 - t|$$

Balio absolutua kontuan hartuz, bi aukera dagoz:

- Balio absolutuen barruko adierazpen biak zeinu berekoak izatea.
- Balio " " " " zeinuz aurkakak " .

$$\begin{array}{l} \swarrow 18t + 1 = 20 - t \rightarrow 19t = 19 \rightarrow \boxed{t_1 = 1} \\ \searrow 18t + 1 = -(20 - t) \rightarrow 17t = -21 \rightarrow \boxed{t_2 = -\frac{21}{17}} \end{array}$$

Beraz, t biak r-ren ekuazioetan ordetkatuz $\Rightarrow$ Bi puntu posible

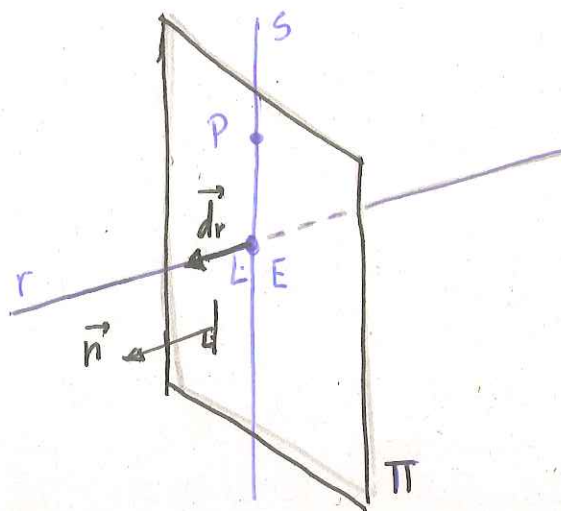
dagoz:

$$\boxed{P_1(4, 2, 4) \quad \text{eta} \quad P_2\left(-\frac{8}{17}, -\frac{80}{17}, -\frac{8}{17}\right)}$$

$$P(1, 0, -2)$$

$$r: \begin{cases} 2x + y - 4z = 7 \\ 2x - y = 5 \end{cases}$$

$$a) \quad s \begin{cases} r \text{ ebaki} \\ s \perp r \\ P \in s \end{cases}$$



- ① r zuzenarekiko perpendikularra den π plano laguntzarilea erabiliko dugu. s zuzena plano horretan dagoen zuzen bat izango da.

$r \perp \pi$ denez $\vec{dr} = \vec{n}$ hartu daiteke. Beraz lehenengo r zuzenaren ekuazio parametrikokoak lortuko duguz ($z=t$) eginez.

$$r: \begin{cases} 2x + y - 4t = 7 \\ 2x - y = 5 \end{cases} \rightarrow \begin{aligned} &2x + 2x - 5 - 4t = 7 \rightarrow 4x = 12 + 4t \\ &\rightarrow y = 2x - 5 \end{aligned} \quad \boxed{x = 3 + t}$$

$$y = 2(3+t) - 5 \rightarrow \boxed{y = 1 + 2t}$$

Beraz,

$$r: \begin{cases} x = 3 + t \\ y = 1 + 2t \\ z = t \end{cases} \Rightarrow \vec{dr} = \vec{n} \underline{\underline{(1, 2, 1)}}$$

π planoaren ekuazio implizitua: $Ax + By + Cz + D = 0$

$$\pi \begin{cases} P(\overset{x}{1}, \overset{y}{0}, \overset{z}{-2}) \\ \vec{n}(\overset{A}{1}, \overset{B}{2}, \overset{C}{1}) \end{cases} \Rightarrow 1 \cdot 1 + 2 \cdot 0 + 1 \cdot (-2) + D = 0 \rightarrow \boxed{D = 1}$$

$$\pi: \boxed{x + 2y + z + 1 = 0}$$

- ② r eta π planoaren arteko ebaki puntua lortuko dugu $\Rightarrow$ (E)
Ekuazio parametrikokoak planoaren ekuazioan ordezkatur:

$$3 + t + 2(1 + 2t) + t + 1 = 0 \rightarrow 6t + 6 = 0 \rightarrow \boxed{t = -1}$$

Ebaki - puntua

$$\begin{cases} x = 3 - 1 \\ y = 1 - 2 \\ z = -1 \end{cases} \Rightarrow \underline{\underline{E(2, -1, -1)}}$$

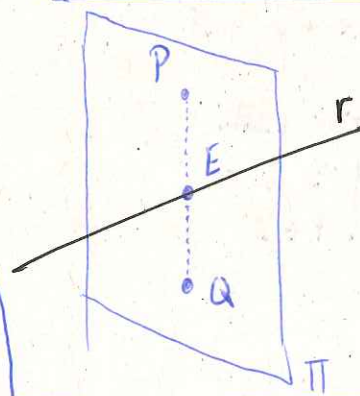
(3) s zuzenaren ekuazioa emen daitekegu:

$$s \begin{cases} P(1, 0, -2) \\ E(2, -1, -1) \end{cases} \rightarrow \vec{d}_s(1, -1, 1)$$

$$s: \left[\frac{x-1}{1} = \frac{y}{-1} = \frac{z+2}{1} \right]$$

b) $\boxed{\text{dist}(P, Q)}$? $\Rightarrow \text{dist}(P, Q) = |\vec{PQ}|$

$Q \rightarrow P$ -ren simetrikoa da r zuzenarekiko



* P eta Q puntuen arteko distantzia eskatzen deusku. Baina simetria dala eta, nahikoa da P eta E-ren arteko distantzia kalkulatzera $\Rightarrow |\vec{PQ}| = 2 \cdot |\vec{PE}|$

Beraz $|\vec{PE}|$ kalkulatuko dugu:

$$\vec{PE} = \vec{d}_s(1, -1, 1)$$

$$|\vec{PE}| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

Beraz, $\boxed{\text{dist}(P, Q) = 2\sqrt{3} \text{ u}}$

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$$A = (1, -1, 0) \quad B(2, 0, 3) \rightarrow \vec{dr} (1, 1, 3)$$

a) $\vec{AB} \perp \pi$?
 $C \in \pi$ $C(2, 2, 3)$

$$r \perp \pi \rightarrow \vec{dr} = \vec{n} (1, 1, 3) \quad \left\{ \begin{array}{l} Ax + By + Cz + D = 0 \rightarrow 1 \cdot 2 + 1 \cdot 2 + 3 \cdot 3 + D = 0 \\ C(2, 2, 3) \end{array} \right.$$

$$D = -13$$

$$\pi: \boxed{x + y + 3z - 13 = 0}$$

b) $\vec{AB} = (1, 1, 3)$
 $\vec{BC} = (0, 2, 0)$ $\left\{ \begin{array}{l} \text{Et dira proportionalak, beraz A, B eta C et dagoz} \\ \text{lerokatu eta ondorioz et dago zuzen bat hiru} \\ \text{puntuetatik pasatzen denik.} \end{array} \right.$

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$$r: \frac{x-2}{a} = \frac{y-1}{4} = \frac{z+1}{2} \rightarrow \vec{dr} (a, 4, 2) \quad P(2, 1, -1)$$

$$\pi: 2x - y + bz = 0 \rightarrow \vec{n} (2, -1, b)$$

a) $r \perp \pi \rightarrow \vec{dr} \parallel \vec{n} \rightarrow$ Beraz proportionalak izan behar dira bektoreak.

$$\frac{a}{2} = \frac{4}{-1} = \frac{2}{b} \rightarrow \boxed{a = -8} \text{ eta } \boxed{b = -\frac{1}{2}}$$



b) $r \in \pi \rightarrow \vec{dr} \perp \vec{n} \rightarrow \boxed{\vec{dr} \cdot \vec{n} = 0}$



$$(a, 4, 2) \cdot (2, -1, b) = 0 \Rightarrow 2a - 4 + 2b = 0 \rightarrow 2a = 4 - 2b$$

$$a = 2 - b$$

$r \in \pi$ bada $\Rightarrow r$ -ko puntu guztiak planean dagoz.

Beraz $P \in \pi \Rightarrow \pi$ -ren ekuazioan ordezkatur:

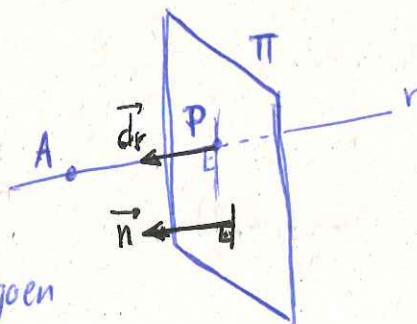
$$2 \cdot 2 - 1 + b \cdot (-1) = 0 \rightarrow 4 - 1 - b = 0 \rightarrow \boxed{b = 3}$$

$$\boxed{a = -1}$$

$$A(2,1,0)$$

$$\Pi: 2x + 3y + 4z = 0$$

a) $\text{dist}(A, \Pi)$



A puntutik distantzia minimora dagoen

Π planoaren P puntua $\rightarrow$ r zutena eta Π planoaren arteko ebaki puntua da ($r \perp \Pi$ izanik).

$$r \perp \Pi \Rightarrow \vec{dr} = \vec{n} (2, 3, 4)$$

Zutena ekuazio parametrikokoak:

$$r: \begin{cases} x = 2 + 2t \\ y = 1 + 3t \\ z = 4t \end{cases}$$

P ebaki puntua lortzeko $\rightarrow$ r-ren ekuazioak Π planoan ordezkatu:

$$2(2+2t) + 3(1+3t) + 4 \cdot 4t = 0 \rightarrow 4 + 4t + 3 + 9t + 16t = 0$$

$$29t = -7 \rightarrow t = -\frac{7}{29}$$

$$P \text{ ebaki-puntua} \Rightarrow \begin{cases} x = 2 + 2 \cdot \frac{-7}{29} \\ y = 1 + 3 \cdot \frac{-7}{29} \\ z = 4 \cdot \frac{-7}{29} \end{cases}$$

$$\Rightarrow P\left(\frac{44}{29}, \frac{8}{29}, \frac{-28}{29}\right)$$

(ondo dago?!)

$$\text{dist}(A, \Pi) = \frac{|2 \cdot 2 + 3 \cdot 1 + 4 \cdot 0|}{\sqrt{2^2 + 3^2 + 4^2}} \Rightarrow \boxed{\text{dist}(A, \Pi) = 1'3 \text{ u}} \left(\text{edo } \frac{7}{\sqrt{29}} \text{ u} \right)$$

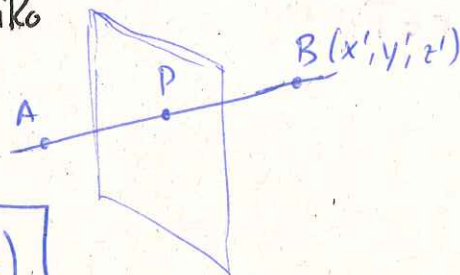
b) B puntua $\Rightarrow$ A-ren simetrikoa Π -rekiko

$$\frac{44}{29} = \frac{2+x'}{2}$$

$$\frac{8}{29} = \frac{1+y'}{2}$$

$$\frac{-28}{29} = \frac{0+z'}{2}$$

$$\Rightarrow \boxed{B\left(\frac{30}{29}, \frac{-13}{29}, \frac{-56}{29}\right)}$$



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$$A(-1, 3, 2)$$

$$B(2, -1, -1)$$

$$C(a-2, 7, b)$$



a) A, B eta C lerokatu ta $\Rightarrow \vec{AB}$ eta $\vec{BC}$ proporzionalak izan behar dira.

$$\vec{AB}(3, -4, -3) \Rightarrow \frac{3}{a-4} = \frac{-4}{8} = \frac{-3}{b+1}$$

$$\vec{BC}(a-4, 8, b+1)$$

$$\frac{3}{a-4} = \frac{-4}{8} \rightarrow 24 = -4a + 16 \rightarrow 4a = -8 \rightarrow \boxed{a = -2}$$

$$\frac{-4}{8} = \frac{-3}{b+1} \rightarrow -4b - 4 = -24 \rightarrow -4 + 24 = 4b \rightarrow \boxed{b = 5}$$

b) $\pi \begin{cases} P(0, -3, 5) \\ \vec{AC} \perp \pi \rightarrow \vec{AC} = \vec{n} \text{ da} \end{cases}$

$$C(-4, 7, 5) \Rightarrow \vec{AC}(-3, 4, 3) = \vec{n}$$

$$A(-1, 3, 2)$$

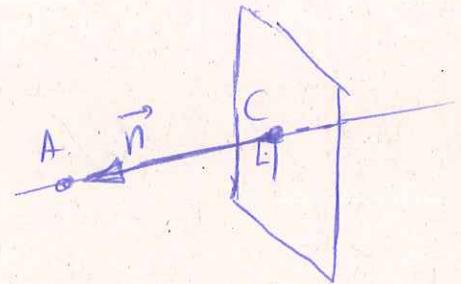
$$\pi: \begin{cases} P(\overset{x}{0}, \overset{y}{-3}, \overset{z}{5}) \\ \vec{n}(\overset{A}{-3}, \overset{B}{4}, \overset{C}{3}) \end{cases}$$

$$Ax + By + Cz + D = 0$$

$$-3 \cdot 0 + 4 \cdot (-3) + 3 \cdot 5 + D = 0 \rightarrow -12 + 15 + D = 0$$

$$\boxed{D = -3}$$

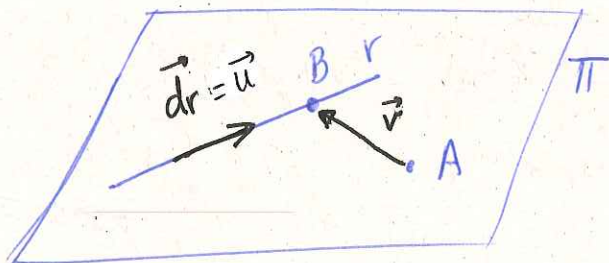
$$\pi: \underline{\underline{-3x + 4y + 3z - 3 = 0}}$$



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$$r: \begin{cases} 3x + 4y + 5z = 0 \\ 2x + y + z = 0 \end{cases}$$

$$A(-1, 2, 1)$$



a) $\Pi \begin{cases} r \in \Pi \\ A \in \Pi \end{cases}$

Lortu daigun r ekuazio parametrikotan $\Rightarrow \boxed{z = t}$

$$r: \begin{cases} 3x + 4y + 5t = 0 \rightarrow 3x + 4(-2x - t) + 5t = 0 \rightarrow 3x - 8x - 4t + 5t = 0 \\ 2x + y + t = 0 \rightarrow y = -2x - t \end{cases}$$

$$-5x + t = 0$$

$$5x = t \rightarrow \boxed{x = \frac{t}{5}}$$

$$y = -2\frac{t}{5} - t \rightarrow y = -\frac{2t}{5} - t$$

$$\boxed{y = -\frac{7t}{5}}$$

$$r: \begin{cases} x = \frac{1}{5}t \\ y = -\frac{7}{5}t \\ z = t \end{cases} \rightarrow \vec{dr} \left(\frac{1}{5}, -\frac{7}{5}, 1 \right) \text{ edo proportionala dan}$$

beste bektore bat hartu ahal dogu:

$$\vec{dr} (1, -7, 5) = \vec{u}$$

Planoaren beste bektore bat behar dogu $\rightarrow$ Adibidez $\vec{AB}$

B puntua r zuzeneko edozein puntu izan daiteke $\rightarrow B(0, 0, 0)$

$$\vec{AB} = \vec{v} (-1, 2, 1)$$

Beraz, Π planoaren ekuazio parametrikak:

$$\Pi: \begin{cases} B(0, 0, 0) \\ \vec{u} \\ \vec{v} \end{cases}$$

$$\Pi: \begin{cases} x = \lambda - \mu \\ y = -7\lambda + 2\mu \\ z = 5\lambda + \mu \end{cases}$$

$$\begin{vmatrix} 1 & -1 & x \\ -7 & 2 & y \\ 5 & 1 & z \end{vmatrix} = 0 \rightarrow$$

$$\rightarrow 2z - 7x - 5y - 10x - y - 7z = 0 \rightarrow \boxed{-17x - 6y - 5z = 0} \text{ edo}$$

$$\boxed{17x + 6y + 5z = 0}$$

$$b) \pi' \begin{cases} B(0, 0, -3) \in \pi' \\ \pi' \parallel \pi \rightarrow \vec{n}' = \vec{n}(17, 6, 5) \end{cases}$$

$$Ax + By + Cz + D = 0$$

$$17 \cdot 0 + 6 \cdot 0 + 5 \cdot (-3) + D = 0 \rightarrow D = 15$$

$$\pi': \boxed{17x + 6y + 5z + 15 = 0}$$

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$$A(1, 2, 3)$$

$$B(1, -2, 4)$$

$$C(1, -3, a)$$

a) 3 puntuak loturkatuta egoteko $\rightarrow \vec{AB}$ eta $\vec{BC}$ proporzionalak izan behar dira:

$$\left. \begin{array}{l} \vec{AB}(0, -4, 1) \\ \vec{BC}(0, -1, a-4) \end{array} \right\} \rightarrow \frac{-4}{-1} = \frac{1}{a-4} \rightarrow \begin{array}{l} -4a + 16 = -1 \\ 16 + 1 = 4a \end{array} \Rightarrow$$

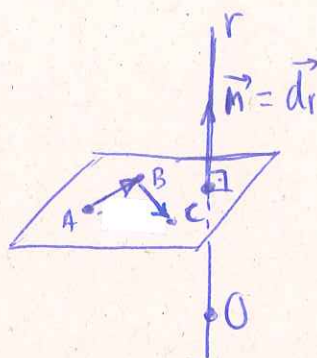
$$\rightarrow \text{Beraz, } \boxed{a = \frac{17}{4}} \text{ izan behar da}$$

$$b) a = 5 \rightarrow C(1, -3, 5)$$

$$O(0, 0, 0) \in r$$

$$r \perp \pi \rightarrow \vec{dr} = \vec{n}$$

$$\pi: \begin{cases} A \\ B \\ C \end{cases}$$



Planoko bi bektoreen biderkadura bektorialetik $\vec{n}$ lortu daiteke.

$$\begin{array}{l} \vec{AB}(0, -4, 1) \\ \vec{BC}(0, -1, 1) \end{array} \left\{ \vec{n} = \vec{dr} = (0, -4, 1) \times (0, -1, 1) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & -4 & 1 \\ 0 & -1 & 1 \end{vmatrix} \Rightarrow \right.$$

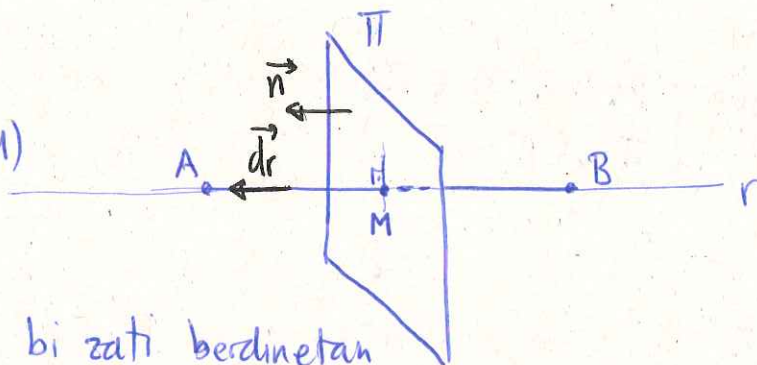
$$\Rightarrow \vec{dr} = -4\vec{j} + \vec{i} = -3\vec{i} \rightarrow \vec{dr}(-3, 0, 0) \text{ (edo } (1, 0, 0))$$

$$\text{Beraz, } \boxed{r: \frac{x}{-3} = \frac{y}{0} = \frac{z}{0}}$$

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$$\Pi: x + y + z = 4 \Rightarrow \vec{n}(1, 1, 1)$$

$$A(1, 0, 0)$$



Planoak AB segmentuan bi zati berdinetan

zatiñen badau $\Rightarrow$ A eta B simetrikoak dira Π planoarekiko.

Segmentuari r zuzena deitzen:

$$r \perp \Pi \rightarrow \vec{dr} = \vec{n}(1, 1, 1)$$

$$r \begin{cases} A(1, 0, 0) \\ \vec{dr}(1, 1, 1) \end{cases} \Rightarrow r: \begin{cases} x = 1 + t \\ y = t \\ z = t \end{cases}$$

[M] Ebaki puntua lortzeko (M) $\Rightarrow$ r-ren parametrizazioa Π planoan ordezkatuz:

$$(1+t) + t + t = 4 \rightarrow 3t = 3 \rightarrow \boxed{t=1}$$

Beraz, M ebaki-puntua: $\boxed{M(2, 1, 1)}$

B puntua: $\left(M = \frac{A+B}{2} \right) \quad B(x, y, z)$

$$\left. \begin{aligned} 2 &= \frac{1+x}{2} \\ 1 &= \frac{0+y}{2} \\ 1 &= \frac{0+z}{2} \end{aligned} \right\} \Rightarrow \boxed{B(3, 2, 2)}$$